



Maintenance Schedule

Diesel-electric drilling unit
PEDPG12V4000-2A0
12V4000G73
Application group 3B

MS50075/03E

System designation	Engine model	Hz	kW	rpm	Application group
PEDPG 12V4000-2A0	12V4000G73	60 Hz	1 105 kW	1200 1/min	3B, continuous operation, variable load, ICXN

Table 1: Applicability

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

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1 Maintenance Schedule

1.1 Preface

Maintenance concept

Information about the latest maintenance schedule is available at: <http://www.mtu-solutions.com>.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of MTU. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (hrs)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the engine is to remain out of service for a longer time period, MTU recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Load profile

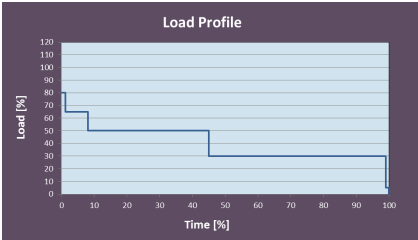
Load profile		Load factor %	Load indicator %	0 to 92.1 kW/cyl.	Load profile			
Load %	Time %							
110 ₍₁₀₀₋₁₁₀₎	—	0 to 40	≤ 1	30,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	—							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	1.3							
65 ₍₅₀₋₆₅₎	6.8							
50 ₍₃₀₋₅₀₎	36.9							
30 ₍₅₋₃₀₎	54.1							
Idle speed ₍₀₋₅₎	0.9							

Table 2: Load profile

1.3 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0010	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.4 Maintenance Schedule Matrix 40000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00321	1 DAY																														
WM00174	1 WK																														
WM00067	2 YR																														
WM00116	2 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00066	9 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00245	3 MON		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00076	3 YR						X						X						X						X						X
WM00664	3 YR						X						X						X						X						X
WM00052	18 YR						X						X						X						X						X
WM00094	18 YR		X				X						X						X						X						X
WM00089	18 YR															X															X
WM00090	18 YR															X															X
WM00098	18 YR															X															X
WM00110	18 YR															X															X
WM00775	18 YR																														X
WM00008	6 YR																														X
WM00065	9 YR																														X
WM00054	18 YR																														X
WM00056	18 YR																														X
WM00062	18 YR																														X
WM00091	18 YR																														X
WM00093	18 YR																														X
WM00096	18 YR																														X
WM00117	18 YR																														X
WM00164	18 YR																														X
WM00921	18 YR																														X
WM00971	18 YR																														X
WM00973	18 YR																														X
WM00974	18 YR																														X
WM00976	18 YR																														X
WM00981	18 YR																														X
WM00986	18 YR																														X
DAY = Days																															
WK = Weeks																															
YR = Years																															
MON = Months																															
HRS = Hours																															

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Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00006	6 YR																														
WM00011	18 YR																														
WM00072	18 YR																														
WM00087	18 YR																														
WM00090	18 YR																														
WM00001	18 YR																														
WM00002	18 YR																														
WM00070	18 YR																														
WM00088	18 YR																														
WM00092	18 YR																														
WM00095	18 YR																														
WM00099	18 YR																														
WM00100	18 YR																														
WM00103	18 YR																														
WM00104	18 YR																														
WM00108	18 YR																														
WM00111	18 YR																														
WM00137	18 YR																														
WM00353	18 YR																														
WM00439	18 YR																														
WM01490	3 MON	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00222	1 YR																														
WM00148	3 YR																														
WM00979	18 YR																														
WM00881	1 MON																														
WM00925	18 YR																														
WM00323	1 DAY																														
WM01476	1 DAY																														
WM01477	1 WK																														
WM00594	6 MON				X				X				X				X				X				X				X		
WM01004	6 MON				X				X				X				X				X				X				X		
WM01478	6 MON				X				X				X				X				X				X				X		
WM01479	6 MON				X				X				X				X				X				X				X		
WM00527	3 YR																				X										
WM01006	3 YR																				X										
WM01480	3 YR																				X										
WM01481	3 YR																				X										
WM01005	3 YR																														
WM01007	18 YR																														
WM00215	1 DAY																														
WM00216	1 DAY																														
WM00214	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM01482	1 DAY																														
WM01483	1 WK																														
DAY = Days																															
WK = Weeks																															
YR = Years																															
MON = Months																															
HRS = Hours																															

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00685	3 MON																														
WM01484	-		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM01485	-						X					X						X							X						X
WM01486	18 YR																														
WM01487	18 YR																														
WM01488	18 YR																														
WM00142	-		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM01489	1 DAY																														
DAY = Days																															
WK = Weeks																															
YR = Years																															
MON = Months																															
HRS = Hours																															

15500 to 30000 Operating hours

Task	Limit	Operating hours [HRS]																													
		15500	16000	16500	17000	17500	18000	18500	19000	19500	20000	20500	21000	21500	22000	22500	23000	23500	24000	24500	25000	25500	26000	26500	27000	27500	28000	28500	29000	29500	30000
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00321	1 DAY																														
WM00174	1 WK																														
WM00067	2 YR																														
WM00116	2 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00066	9 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00245	3 MON		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00076	3 YR						X						X						X						X						X
WM00664	3 YR						X						X						X						X						X
WM00052	18 YR						X						X						X						X						X
WM00094	18 YR						X						X						X						X						X
WM00089	18 YR															X															X
WM00090	18 YR															X															X
WM00098	18 YR															X															X
WM00110	18 YR															X															X
WM00775	18 YR																														X
WM00008	6 YR																														X
WM00065	9 YR																														X
WM00054	18 YR																														X
DAY = Days WK = Weeks YR = Years MON = Months HRS = Hours																															

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Task	Limit	Operating hours [HRS]																													
		15500	16000	16500	17000	17500	18000	18500	19000	19500	20000	20500	21000	21500	22000	22500	23000	23500	24000	24500	25000	25500	26000	26500	27000	27500	28000	28500	29000	29500	30000
WM00056	18 YR																														X
WM00062	18 YR																														X
WM00091	18 YR																														X
WM00093	18 YR																														X
WM00096	18 YR																														X
WM00117	18 YR																														X
WM00164	18 YR																														X
WM00921	18 YR																														X
WM00971	18 YR																														X
WM00973	18 YR																														X
WM00974	18 YR																														X
WM00976	18 YR																														X
WM00981	18 YR																														X
WM00986	18 YR																														X
WM00006	6 YR																														X
WM00011	18 YR																														X
WM00072	18 YR																														X
WM00087	18 YR																														X
WM00980	18 YR																														X
WM00001	18 YR																														X
WM00002	18 YR																														X
WM00070	18 YR																														X
WM00088	18 YR																														X
WM00092	18 YR																														X
WM00095	18 YR																														X
WM00099	18 YR																														X
WM00100	18 YR																														X
WM00103	18 YR																														X
WM00104	18 YR																														X
WM00108	18 YR																														X
WM00111	18 YR																														X
WM00137	18 YR																														X
WM00353	18 YR																														X
WM00439	18 YR																														X
WM01490	3 MON	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00222	1 YR																														
WM00148	3 YR																														
WM00979	18 YR																														X
WM00881	1 MON																														
WM00925	18 YR																														X
WM00323	1 DAY																														
WM01476	1 DAY																														
WM01477	1 WK																														
WM00594	6 MON		X				X				X				X				X				X				X				X
DAY = Days																															
WK = Weeks																															
YR = Years																															
MON = Months																															
HRS = Hours																															

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Task	Limit	Operating hours [HRS]																															
		15500	16000	16500	17000	17500	18000	18500	19000	19500	20000	20500	21000	21500	22000	22500	23000	23500	24000	24500	25000	25500	26000	26500	27000	27500	28000	28500	29000	29500	30000		
WM01004	6 MON		X				X				X				X				X				X				X					X	
WM01478	6 MON		X				X				X				X				X				X				X					X	
WM01479	6 MON		X				X				X				X				X				X				X					X	
WM00527	3 YR										X																					X	
WM01006	3 YR										X																					X	
WM01480	3 YR										X																					X	
WM01481	3 YR										X																					X	
WM01005	3 YR										X																						
WM01007	18 YR																																
WM00215	1 DAY																																
WM00216	1 DAY																																
WM00214	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM01482	1 DAY																																
WM01483	1 WK																																
WM00685	3 MON																																
WM01484	-		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM01485	-						X					X					X						X									X	
WM01486	18 YR																																X
WM01487	18 YR																																X
WM01488	18 YR																																X
WM00142	-		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM01489	1 DAY																																
DAY = Days																																	
WK = Weeks																																	
YR = Years																																	
MON = Months																																	
HRS = Hours																																	

30500 to 40000 Operating hours

Task	Limit	Operating hours [HRS]																													
		30500	31000	31500	32000	32500	33000	33500	34000	34500	35000	35500	36000	36500	37000	37500	38000	38500	39000	39500	40000										
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00321	1 DAY																														
WM00174	1 WK																														
WM00067	2 YR																														
WM00116	2 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X										
WM00066	9 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X										
WM00245	3 MON		X		X		X		X		X		X		X		X		X		X										
DAY = Days																															
WK = Weeks																															
YR = Years																															
MON = Months																															
HRS = Hours																															

Task	Limit	Operating hours [HRS]																							
		30500	31000	31500	32000	32500	33000	33500	34000	34500	35000	35500	36000	36500	37000	37500	38000	38500	39000	39500	40000				
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X				
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X				
WM00076	3 YR						X						X						X						
WM00664	3 YR						X						X						X						
WM00052	18 YR						X						X						X						
WM00094	18 YR						X						X						X						
WM00089	18 YR																X								
WM00090	18 YR															X									
WM00098	18 YR															X									
WM00110	18 YR															X									
WM00775	18 YR																								
WM00008	6 YR																								
WM00065	9 YR																								
WM00054	18 YR																								
WM00056	18 YR																								
WM00062	18 YR																								
WM00091	18 YR																								
WM00093	18 YR																								
WM00096	18 YR																								
WM00117	18 YR																								
WM00164	18 YR																								
WM00921	18 YR																								
WM00971	18 YR																								
WM00973	18 YR																								
WM00974	18 YR																								
WM00976	18 YR																								
WM00981	18 YR																								
WM00986	18 YR																								
WM00006	6 YR																								
WM00011	18 YR																								
WM00072	18 YR																								
WM00087	18 YR																								
WM00980	18 YR																								
WM00001	18 YR																								
WM00002	18 YR																								
WM00070	18 YR																								
WM00088	18 YR																								
WM00092	18 YR																								
WM00095	18 YR																								
WM00099	18 YR																								
WM00100	18 YR																								
WM00103	18 YR																								
WM00104	18 YR																								
WM00108	18 YR																								
DAY = Days																									
WK = Weeks																									
YR = Years																									
MON = Months																									
HRS = Hours																									

[illegible]

1.5 Cyclical maintenance tasks 40000 HRS

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL 1	-	1 DAY	ENGINE MONITORING	Check engine oil level		WM00285
QL 1	-	1 DAY	ENGINE MONITORING	Check general condition of engine and inspect for leaks		WM00286
QL 1	-	1 DAY	ENGINE MONITORING	Check maintenance indicator of air filter	X	WM00287
QL 1	-	1 DAY	ENGINE MONITORING	Check relief bore of coolant pump		WM00288
QL 1	-	1 DAY	ENGINE MONITORING	When engine is running, check for abnormal running noises, exhaust gas color and vibrations		WM00289
QL 1	-	1 DAY	ENGINE MONITORING	Check condensate drain of intercooler		WM00321
QL 1	-	1 WK	EMERGENCY AIR SHUT-OFF FLAPS	Check operation of emergency air shut-off flaps	X	WM00174
QL 1	-	2 YR	LUBE OIL SYSTEM	Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest	X	WM00067
QL 1	-	2 YR	COOLANT SYSTEM	Replace filter		WM00116
QL 1	500	2 YR	LUBE OIL SYSTEM	Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest	X	WM00063
QL 1	500	9 YR	AUTOMATIC OIL FILTER	Automatic oil filter, check and clean filtering screen	X	WM00066
QL 1	1000	3 MON	EMERGENCY AIR SHUT-OFF FLAPS	Lubricate lubrication points of emergency air shut-off flaps		WM00245
QL 1	1000	2 YR	FUEL SYSTEM (LOW PRES-SURE)	Replace easy-change fuel filter		WM00086
QL 1	1000	2 YR	AUXILIARY POWER TAKE-OFF	Check belt condition, replace if necessary. Adjust tension.		WM00170
QL 1	3000	3 YR	AIR SUPPLY SYSTEM	Replace air filter	X	WM00076
QL 1	3000	3 YR	COOLANT SYSTEM	Replace valve cover	X	WM00664
QL 1	3000	18 YR	BATTERY-CHARGING GENERATOR	Check condition of coupling	X	WM00052
QL 1	3000	18 YR	VALVE GEAR	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		WM00094
QL 1	7500	18 YR	GOVERNOR	Reset drift correction parameters (CDC)		WM00089
QL 1	7500	18 YR	FUEL SYSTEM (HIGH PRES-SURE)	Replace fuel injection valve/injector		WM00090
QL 1	7500	18 YR	CYLINDER HEAD	Measure valve protrusion		WM00098
QL 1	7500	18 YR	RUNNING GEAR	Inspect cylinder liner with endoscope		WM00110
QL 1	15000	18 YR	CRANKCASE BREATHER	Replace oil separator (cyclone)		WM00775
QL 3	15000	6 YR	RUBBER SLEEVE	Replace rubber sleeve		WM00008
QL 3	15000	9 YR	AUTOMATIC OIL FILTER	Automatic oil filter, remove and clean filter housing, replace filter element and bearing	X	WM00065
QL 3	15000	18 YR	COOLANT SYSTEM	Check LT thermostat housing and replace thermal actuator		WM00054
QL 3	15000	18 YR	COOLANT SYSTEM	Check HT thermostat housing and replace thermal actuator		WM00056
QL 3	15000	18 YR	LUBE OIL SYSTEM	Check centrifugal oil filter for wear	X	WM00062
QL 3	15000	18 YR	FUEL SYSTEM (HIGH PRES-SURE)	Replace HP pump		WM00091
QL 3	15000	18 YR	VALVE GEAR	Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore		WM00093
QL 3	15000	18 YR	VALVE GEAR	Visually inspect camshaft running surfaces with endoscope through pushrod bore		WM00096
QL 3	15000	18 YR	EXHAUST TURBOCHARGER	Replace exhaust turbocharger		WM00117

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TIM-ID: 0000000000 - 000

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL3	15000	18 YR	AIR PIPING TO CYLINDERS	Clean air piping		WM00164
QL3	15000	18 YR	COOLANT SYSTEM	Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)		WM00921
QL3	15000	18 YR	STARTING SYSTEM	Overhaul or replace starter		WM00971
QL3	15000	18 YR	COOLANT SYSTEM	Overhaul or replace coolant pump in LT circuit		WM00973
QL3	15000	18 YR	COOLANT SYSTEM	Overhaul or replace coolant pump in HT circuit		WM00974
QL3	15000	18 YR	BATTERY-CHARGING GENERATOR	Overhaul or replace battery-charging generator	X	WM00976
QL3	15000	18 YR	LUBE OIL SYSTEM	Clean oil cooler and check for leaks or replace		WM00981
QL3	15000	18 YR	AIR PIPING TO CYLINDERS	Clean intercooler and check for leaks or replace		WM00986
QL3	30000	6 YR	HOSE	Replace hose line		WM00006
QL3	30000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP fuel sensor		WM00011
QL3	30000	18 YR	EXHAUST SYSTEM	Replace exhaust pipe bellows		WM00072
QL3	30000	18 YR	FUEL SYSTEM (LOW PRESSURE)	Replace pump		WM00087
QL3	30000	18 YR	CYLINDER HEAD	Overhaul or replace cylinder head		WM00980
QL4	30000	18 YR	ENGINE, GENERAL	Replace elastomer parts and seals		WM00001
QL4	30000	18 YR	ENGINE, GENERAL	Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary.		WM00002
QL4	30000	18 YR	LUBE OIL SYSTEM	Check pump, replace if necessary		WM00070
QL4	30000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace pressure relief valve		WM00088
QL4	30000	18 YR	VALVE GEAR	Replace swing follower and swing follower shaft		WM00092
QL4	30000	18 YR	VALVE GEAR	Replace thrust bearing flange		WM00095
QL4	30000	18 YR	RUNNING GEAR	Replace conrod bearings		WM00099
QL4	30000	18 YR	RUNNING GEAR	Replace rings		WM00100
QL4	30000	18 YR	RUNNING GEAR	Replace crankshaft bearings		WM00103
QL4	30000	18 YR	GEAR TRAIN	Check tooth flanks in gear train for wear (visual inspection), replace bushing		WM00104
QL4	30000	18 YR	RUNNING GEAR	Replace cylinder liners		WM00108
QL4	30000	18 YR	CAMSHAFT	Replace camshaft bearings		WM00111
QL4	30000	18 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace wiring harness		WM00137
QL4	30000	18 YR	RUNNING GEAR	Replace conrod		WM00353
QL4	30000	18 YR	RUNNING GEAR	Replace vibration damper		WM00439
Starter						
QL1	500	3 MON	STARTING SYSTEM	Check acid level in starting battery, top up if required. Clean poles if necessary		WM01490
Preheater						
QL1	-	1 YR	PREHEATING	Preheater, check for function and leaks	X	WM00222
QL1	-	3 YR	PREHEATING	Preheater, replace thermostat (electric)	X	WM00148
QL1	30000	18 YR	PREHEATING	Overhaul or replace preheating unit	X	WM00979
Coupling						
QL1	-	1 MON	BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING	Check coupling (visual inspection), replace as necessary		WM00881
QL4	30000	18 YR	BENDING / TORSIONALLY RESILIENT COUPLING	Replace coupling or coupling element(s)		WM00925

Generator

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Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL 1	-	1 DAY	GENERATOR	Check general condition (visual inspection)		WM00323
QL 1	-	1 DAY	GENERATOR	Inspect bearing housing for traces of grease and/or oil		WM01476
QL 1	-	1 WK	GENERATOR	Check dust filter for contamination		WM01477
QL 1	2000	6 MON	GENERATOR	Check insulation resistance		WM00594
QL 1	2000	6 MON	GENERATOR	Clean interior of junction box, air grilles and air deflectors with compressed air or, if necessary, with a solvent suitable for electric components		WM01004
QL 1	2000	6 MON	GENERATOR	Clean terminal box, cable connection, terminals of protective and grounding conductors, contacts, and check for firm seating		WM01478
QL 1	2000	6 MON	GENERATOR	Check for abnormal running noises and vibrations, drain condensate		WM01479
QL 3	10000	3 YR	GENERATOR	Check function of anti-condensation heating		WM00527
QL 3	10000	3 YR	GENERATOR	Visually inspect fan and fan hub for damage		WM01006
QL 3	10000	3 YR	GENERATOR	Check insulation resistance of stator winding		WM01480
QL 3	10000	3 YR	GENERATOR	Check connections of rotating rectifier for secure seating		WM01481
QL 3	20000	3 YR	GENERATOR	Remove end brackets and visually inspect generator end windings for oil or dirt contamination		WM01005
QL 4	40000	18 YR	GENERATOR	Overhaul generator, check and clean components, and replace diodes and bearings		WM01007
Fuel prefilter						
QL 1	-	1 DAY	FUEL SYSTEM (LOW PRESSURE)	Drain water and contaminants from fuel prefilter	X	WM00215
QL 1	-	1 DAY	FUEL SYSTEM (LOW PRESSURE)	Check position of contamination indicator of fuel prefilter	X	WM00216
QL 1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Replace filter element	X	WM00214
Radiator						
QL 1	-	1 DAY	COOLING SYSTEM	Check temperature of drive belt	X	WM01482
QL 1	-	1 WK	COOLING SYSTEM	Relubricate pedestal bearing. Different grease types must not be mixed		WM01483
QL 1	-	3 MON	COOLANT SYSTEM	Cooler, electric motor - Lubricate bearing	X	WM00685
QL 1	1000	-	COOLING SYSTEM	Visually inspect fan guard. Re-tighten connections if necessary		WM01484
QL 1	3000	-	COOLING SYSTEM	Check condition of drive belt, replace if necessary. Adjust tension. ATTENTION! Re-tension new belt after 24h engine runtime		WM01485
QL 4	30000	18 YR	COOLING SYSTEM	Replace bearings in belt pulleys of fans		WM01486
QL 4	30000	18 YR	COOLING SYSTEM	Overhaul electric motors	X	WM01487
QL 4	30000	18 YR	COOLING SYSTEM	Clean, flush and top up cooling system		WM01488
Control system						
QL 1	1000	-	MONITORING SYSTEM	Visually inspect dampers, replace if necessary		WM00142
Combustion air ducting						
QL 1	-	1 DAY	AIR SUPPLY SYSTEM	Check correct seating of rain cap	X	WM01489
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